DSA0603 – Data Handling and Visualization

CO1 – Assessment Test 3 (AT3): Visualization Design Exercise

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Question 1 – Multi-variable Retail Sales Visualization

Question: A multinational retail company has collected annual sales information from its branches across different regions. Design an effective visualization that simultaneously represents Sales, Profit, Number of Customers, Product Category, and Region. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every design choice.

R Program (ggplot2)

```
library(ggplot2)

retail <- data.frame(
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
           "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
  Region=c("North","South","East","West","North","South","East","West",
           "North","South","East","West","North","South","East","West",
           "North","South","East","West"),
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",
             "Furniture","Clothing","Food","Electronics","Furniture",
             "Clothing","Food","Electronics","Furniture","Clothing",
             "Food","Electronics","Furniture","Clothing","Food"),
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
          560000,670000,195000,980000,430000,760000,280000,1150000,
          510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
           72000,98000,-8000,205000,51000,132000,24000,250000,
           69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
              2100,3300,1100,4700,1750,3600,1450,5600,
              2200,3900,1600,6100)
)

ggplot(retail, aes(x = Sales, y = Profit)) +
  geom_point(aes(size = Customers, color = Category, shape = Region),
             alpha = 0.8, stroke = 0.6) +
  scale_x_log10(labels = scales::comma) +          # log transform: Sales is right-skewed
  scale_size_continuous(range = c(3, 14)) +
  scale_color_brewer(palette = 'Set2') +
  geom_hline(yintercept = 0, linetype = 'dashed', color = 'grey40') +
  labs(title = 'Retail Branch Performance: Sales vs Profit',
       subtitle = 'Bubble size = Customers | Colour = Category | Shape = Region',
       x = 'Sales (log scale)', y = 'Profit') +
  theme_minimal(base_size = 12)
```

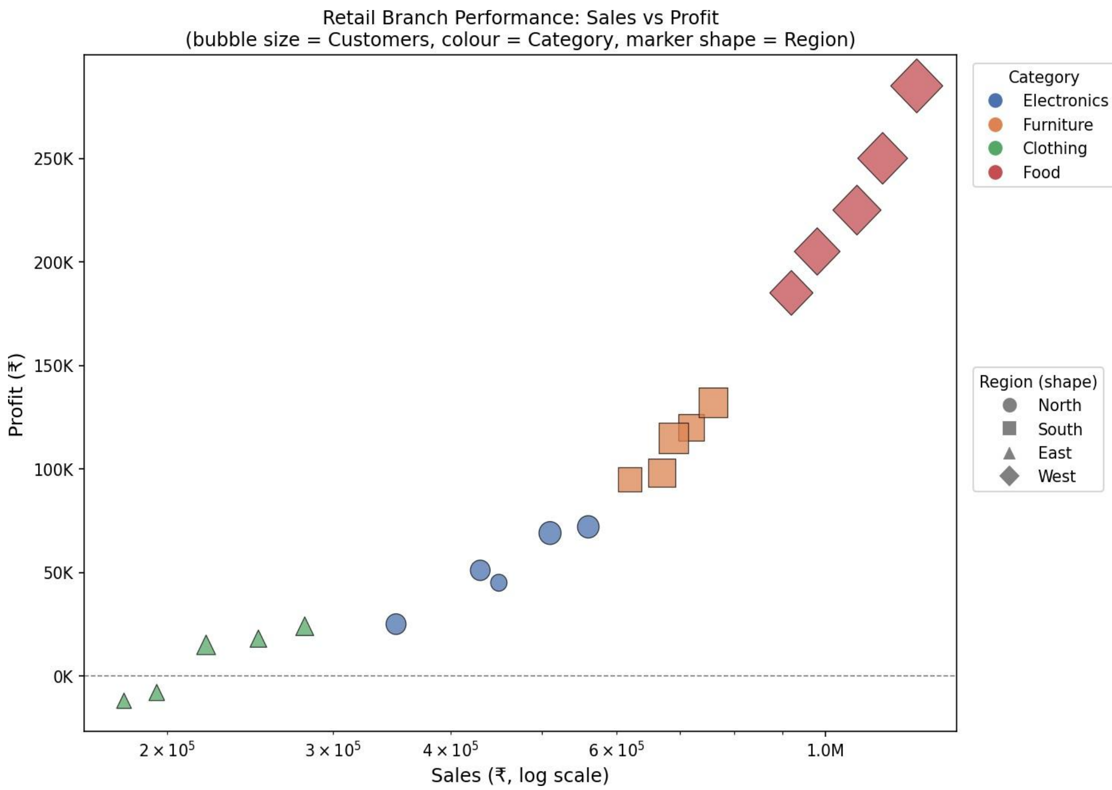

Figure 1: Bubble chart of Sales vs Profit — size encodes Customers, colour encodes Category, marker shape encodes Region.

Design Justification & Interpretation

- Position (x, y): Sales and Profit are the two most important continuous KPIs, so they are mapped to the strongest perceptual channel — position on a common scale (Cleveland & McGill's accuracy ranking).
- Size → Customers: bubble area encodes a third continuous variable (Customers). It is a weaker channel than position, which is appropriate since Customers is a secondary/contextual metric here.
- Colour → Category: a categorical (nominal) qualitative palette (ColorBrewer 'Set2') distinguishes the 4 product categories without implying order.
- Shape → Region: a fifth categorical variable is encoded using marker shape, keeping the chart to a single panel instead of faceting into 4 separate plots, which would make cross-region comparison harder.
- Scale transformation: Sales ranges from ₹1.8 lakh to ₹12.5 lakh (~7x spread) and is right-skewed. A log10 transform on the x-axis spreads out the lower-value branches, prevents high-Sales points from dominating the plot, and makes multiplicative (percentage) differences visually comparable.
- Reference line: a dashed line at Profit = 0 immediately flags loss-making branches (Clothing shows negative profit consistently), which is an important business insight.

Interpretation: Food and Electronics branches dominate high Sales and high Profit with large customer bases (large bubbles), especially in the West region. Clothing branches consistently sit near or below the zero-profit line despite moderate sales, indicating a systemic margin problem for that category across all regions — a pattern that a single-channel table would not reveal as quickly.

Question 2 – Cartesian vs Polar Coordinates

Question: The dataset contains monthly electricity generation from different energy sources. Develop visualizations using Cartesian and Polar coordinates. Compare both and determine which provides better analytical insight. Explain your design decisions.

R Program (ggplot2)

```
library(ggplot2)
library(tidyr)

energy <- data.frame(
  Month=month.abb,
  Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
  Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
  Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
  Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
  Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
)
energy$Month <- factor(energy$Month, levels = month.abb)
long <- pivot_longer(energy, -Month, names_to='Source', values_to='GWh')

# --- Cartesian version ---
ggplot(long, aes(Month, GWh, color = Source, group = Source)) +
  geom_line(linewidth = 1) + geom_point(size = 1.5) +
  labs(title = 'Monthly Generation — Cartesian Coordinates') +
  theme_minimal()

# --- Polar version ---
ggplot(long, aes(Month, GWh, color = Source, group = Source)) +
  geom_line(linewidth = 1) + geom_point(size = 1.5) +
  coord_polar() +
  labs(title = 'Monthly Generation — Polar Coordinates') +
  theme_minimal()
```

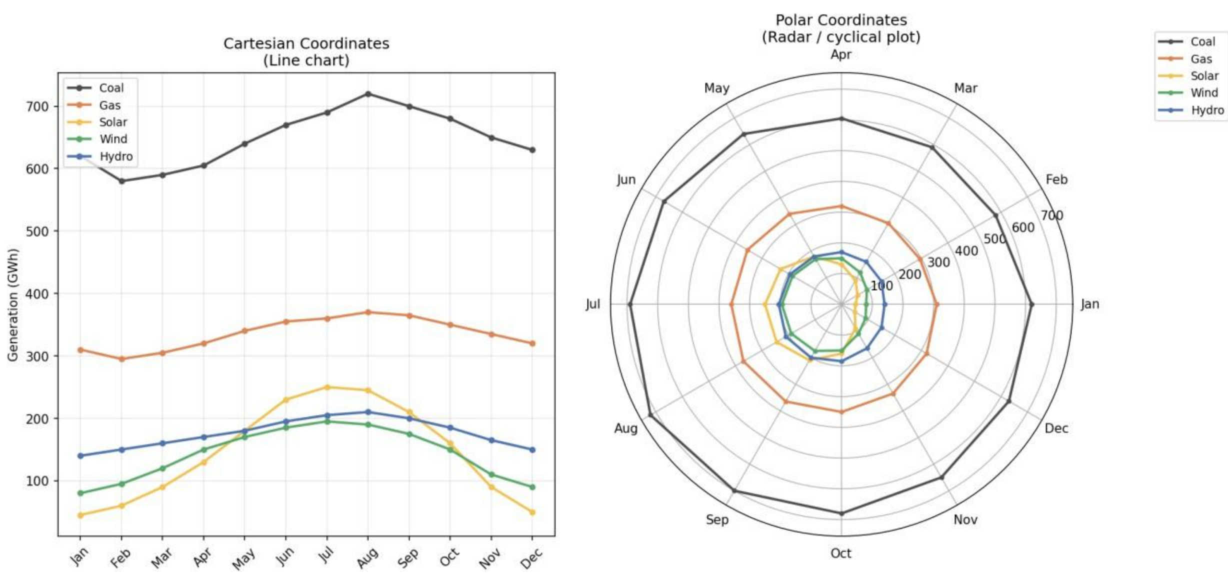

Figure 2: Same data rendered in Cartesian (left) vs Polar (right) coordinates.

Comparison & Justification

- Cartesian: position along a straight, equal-interval x-axis makes it easy to read exact magnitude at any month and to compare slopes (rate of change) between sources. Trends (e.g., Solar's summer peak) and precise cross-source comparison at a given month are unambiguous.
- Polar: wraps the same 12 months around a circle, which naturally represents the cyclical (seasonal) nature of the calendar — December connects back to January. It is visually engaging and good for showing 'shape' of seasonality at a glance.
- Analytical insight: Cartesian coordinates provide better analytical insight for this dataset. Radial distance is harder to judge accurately than linear position (per Cleveland & McGill), curved gridlines distort area/length comparisons, and with 5 overlapping series the polar chart becomes visually cluttered near the centre.

Conclusion: The polar chart is a reasonable choice only for emphasizing cyclical/seasonal identity in a presentation context; for accurate quantitative analysis and comparison across sources, the Cartesian line chart is superior and should be the primary analytical visualization.

Question 3 – Colour Scales for Climate Data

Question: An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents Temperature, AQI, Rainfall, and Humidity using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

R Program (ggplot2)

```
library(ggplot2)

climate <- data.frame(
  City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
         "Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
         "Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
         "Surat","Visakhapatnam","Coimbatore","Mysuru"),
  Temperature=c(44,35,38,40,39,31,33,43,45,39,
                38,40,32,30,41,37,36,34,29,28),
  Humidity=c(22,84,78,72,48,66,58,26,18,42,36,45,89,91,30,34,76,82,68,64),
  Rainfall=c(6,280,220,190,85,120,95,2,1,35,
             48,72,360,410,12,20,180,240,130,145),
  AQI=c(365,110,125,180,160,72,84,290,320,240,
        175,220,58,52,270,190,145,98,60,55)
)

ggplot(climate, aes(Temperature, Humidity)) +
  geom_point(aes(size = Rainfall, color = AQI), alpha = 0.85) +
  ggrepel::geom_text_repel(aes(label = City), size = 3) +
  scale_color_viridis_c(option = 'viridis', direction = -1) +   # colour-blind safe
sequential
  scale_size_continuous(range = c(3, 16)) +
  labs(title = 'Indian Cities: Temperature vs Humidity',
       subtitle = 'Colour = AQI (viridis) | Size = Rainfall',
       color = 'AQI') +
  theme_minimal(base_size = 12)
```

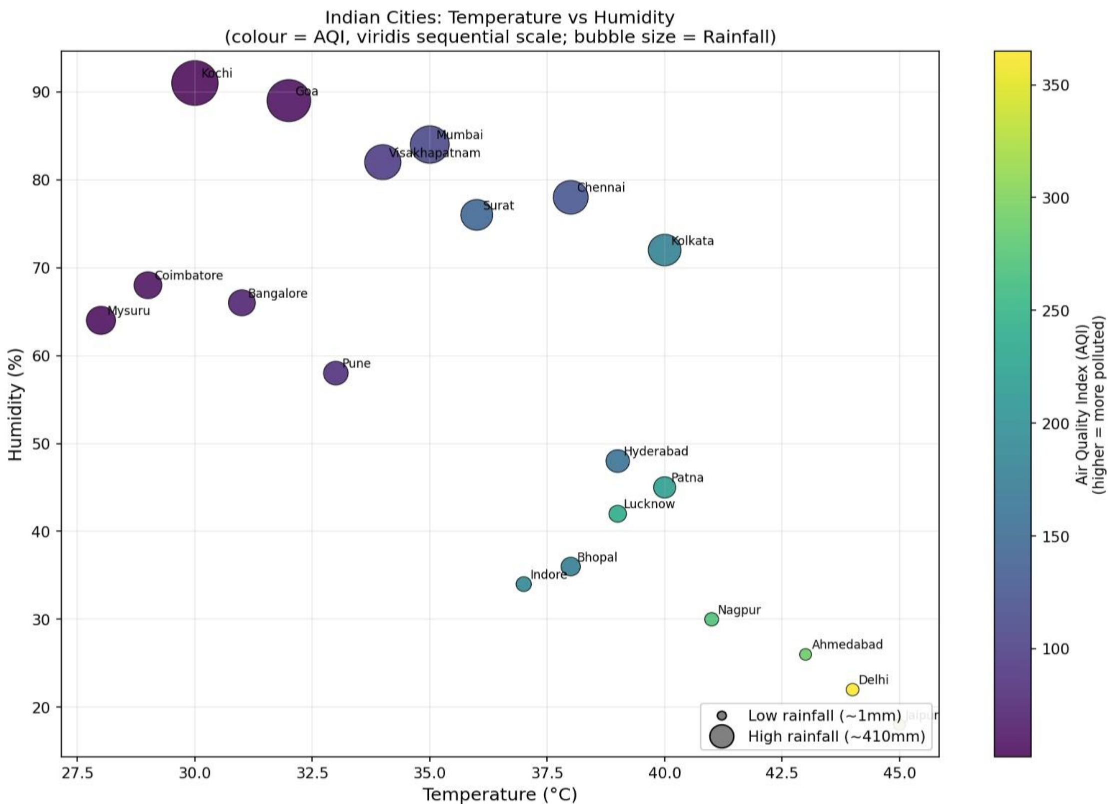

Figure 3: Temperature vs Humidity scatter — colour encodes AQI on a sequential viridis scale, size encodes Rainfall.

Colour Palette Justification

- Palette choice — Viridis (sequential): AQI is a continuous, ordered variable (low → high pollution), so a perceptually-uniform sequential palette is correct — equal numeric differences in AQI produce equal perceived colour differences, unlike rainbow/jet palettes which introduce false banding.
- Accessibility: Viridis is specifically engineered to remain distinguishable for the most common forms of colour-vision deficiency (deuteranopia/protanopia) and prints correctly in grayscale, satisfying colour-blind-safe design guidelines.
- Why not diverging or qualitative: a diverging palette (e.g., red-blue) implies a meaningful midpoint/zero, which AQI does not have in this framing; a qualitative palette would be wrong because AQI is ordinal/continuous, not categorical.
- Position + size were reserved for Temperature/Humidity (x,y — strongest channel) and Rainfall (size — third variable), so all four requested variables are encoded without overloading colour.

Interpretation: Delhi and Jaipur stand out with high temperature, low humidity, and the worst AQI (dark end of the scale) — consistent with dry, polluted northern-plains climates. Coastal cities (Mumbai, Kochi, Goa) show high humidity, high rainfall (large bubbles) and comparatively better AQI, illustrating how monsoon exposure correlates with air quality.

Question 4 – Nonlinear Axis Transformation for Skewed Data

Question: A software company recorded project statistics for twenty development teams. Design a visualization requiring a nonlinear axis transformation to represent highly skewed data. Justify the transformation and explain how it improves interpretation over a linear scale.

R Program (ggplot2)

```
library(ggplot2)
library(patchwork)

software <- data.frame(
  Team=paste("T",1:20,sep=""),
  LinesOfCode=c(12000,18000,25000,35000,48000,
                62000,81000,96000,120000,150000,
                185000,240000,310000,390000,480000,
                620000,810000,1050000,1350000,1800000),
  Bugs=c(35,52,60,72,88,110,125,142,165,188,
         205,240,280,320,365,420,490,560,640,720),
  Developers=c(5,7,9,10,12,14,16,18,20,22,
               25,28,30,34,38,42,46,50,55,60),
  Complexity=c(2.1,2.4,2.7,3.0,3.2,3.5,3.8,4.1,4.5,4.8,
               5.1,5.4,5.8,6.2,6.6,7.1,7.6,8.0,8.5,9.0)
)

p1 <- ggplot(software, aes(LinesOfCode, Bugs, size=Developers, color=Complexity)) +
  geom_point(alpha=0.85) + scale_color_viridis_c(option='plasma') +
  labs(title='Linear Scale') + theme_minimal()

p2 <- ggplot(software, aes(LinesOfCode, Bugs, size=Developers, color=Complexity)) +
  geom_point(alpha=0.85) + scale_color_viridis_c(option='plasma') +
  scale_x_log10(labels = scales::comma) +   # nonlinear transform
  labs(title='Log10 Scale') + theme_minimal()

p1 + p2
```

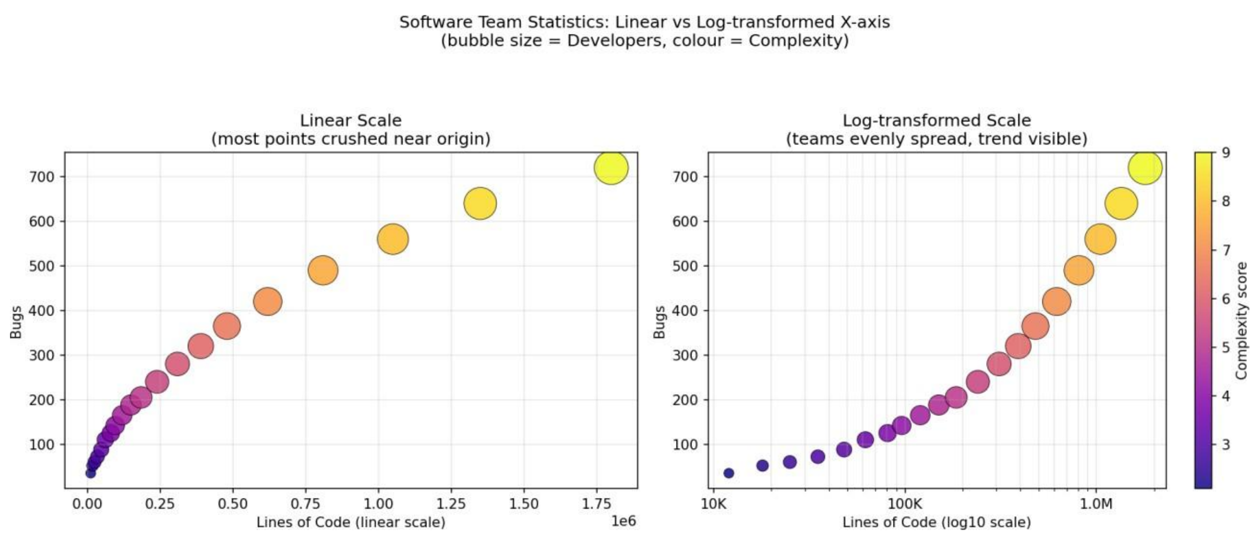

Figure 4: Lines-of-Code vs Bugs on a linear x-axis (left) versus log10-transformed x-axis (right).

Justification of the Transformation

- Skew: LinesOfCode spans 12,000 to 1,800,000 — a 150x range. On a linear axis, 16 of the 20 teams collapse into the leftmost ~10% of the plot, making it impossible to distinguish small/medium teams or see any trend among them.

- Log10 transform: converts multiplicative differences into equal visual distances. Doubling LinesOfCode always moves a point the same distance right, regardless of whether it's 20K→40K or 800K→1.6M — this evenly spreads all 20 teams across the plot.
- Improved interpretation: with the log scale, a clear near-linear relationship between log(LinesOfCode) and Bugs emerges, revealing that bug count grows roughly proportionally with codebase size on a percentage basis — a power-law-like relationship that the linear plot completely hides.
- Colour and size remain unchanged (Complexity, Developers) — only the axis transform changes, isolating the effect of the scale choice for a fair before/after comparison.

Interpretation: On the log-scaled plot, teams with the largest, most complex codebases (T16–T20, darker/larger points) consistently show the highest bug counts and developer headcounts, confirming that code scale and complexity are the dominant drivers of defect volume — a relationship the linear-scale chart could not communicate.

Question 5 – Integrated Company Performance Visualization

Question: A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization communicating Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector, using appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

R Program (ggplot2)

```
library(ggplot2)

companies <- data.frame(
  Company=paste("C",1:20,sep=""),
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy"),
  Revenue=c(420,580,720,850,980,1150,1320,1480,1650,1820,
            2050,2280,2550,2840,3150,3480,3860,4250,4680,5200),
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,20.4,17.8,14.2,10.5,22.1,
                 24.5,19.8,16.7,13.4,26.2,28.1,21.5,18.3,15.2,30.4),
  Employees=c(1500,3200,2100,4200,2800,5100,6800,3900,7200,4500,
              8100,9500,6200,10300,7200,11800,13200,8600,14500,16000),
  MarketCap=c(8500,12000,9800,7600,14500,18200,21500,16800,13200,24500,
              28500,32500,24800,19200,38200,42500,46800,35600,29800,52000)
)

ggplot(companies, aes(Revenue, ProfitMargin)) +
  geom_point(aes(size = MarketCap, color = Sector), alpha = 0.8, stroke = 0.6) +
  scale_x_log10(labels = scales::comma) +          # Revenue spans 420 -> 5200 (12x,
skewed)
  scale_size_continuous(range = c(3, 16)) +
  scale_color_brewer(palette = 'Set2') +           # qualitative, colour-blind friendly
  labs(title = 'Company Performance: Revenue vs Profit Margin',
       subtitle = 'Bubble size = Market Cap | Colour = Sector (log-x)',
       x = 'Revenue (log scale)', y = 'Profit Margin (%)') +
  theme_minimal(base_size = 12)
```

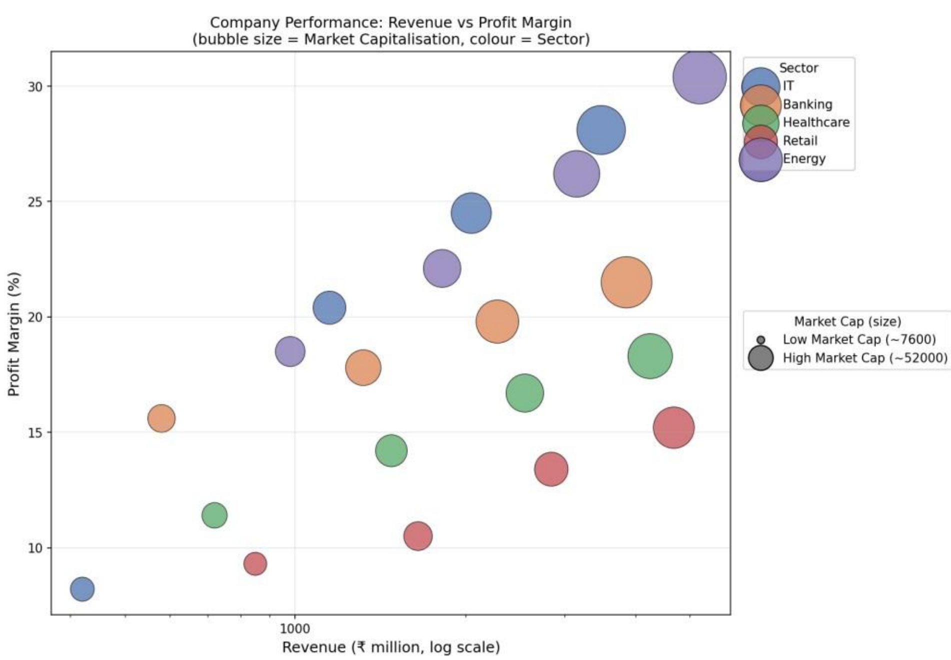

Figure 5: Revenue vs Profit Margin — bubble size encodes Market Capitalisation, colour encodes Sector.

Design Justification & Interpretation:

- Position → Revenue & Profit Margin: the two core financial KPIs use the most accurate channel (position on aligned axes).
- Size → Market Capitalization: encodes a third continuous variable via area; deliberately scaled with sqrt-proportional area (not radius) to avoid the classic perceptual bias where bubble area is over-read when radius is scaled linearly.
- Colour → Sector: a 5-category qualitative ColorBrewer palette ('Set2') is used since Sector is nominal — colours are equally distinguishable with no implied ranking, and the palette is designed to be robust for common colour-vision deficiencies.
- Log scale on Revenue: Revenue spans ₹420M to ₹5,200M (~12x, right-skewed across the 20 companies), so a log10 x-axis prevents the largest companies from dominating and lets percentage-scale differences between small and large firms be compared fairly.
- Employee Strength was deliberately not force-mapped to a 6th channel; adding a 6th aesthetic (e.g., transparency or a secondary shape) would increase cognitive load and risk misinterpretation — instead it is reported separately as supporting context, following the principle of minimizing perceptual bias over maximizing channel count.

Interpretation: Energy and IT companies occupy the upper-right of the chart — highest revenue and the highest profit margins with large market caps, indicating strong scale efficiency. Retail firms cluster at the lower end of profit margin despite moderate-to-high revenue, suggesting thinner margins typical of the sector, while Banking and Healthcare occupy the middle band with steadily increasing margins as revenue grows.